YEAR 12 MATHEMATICS EXTENSION 1

Chapter 11 Trigonometric Equations

Lesson and Homework Pack

Revision	Compound angles, double angles and the t-formulae	Year 11 Extension 1
Lesson 11A	Equations involving compound angles	Past papers: 19 marks · 30 min
Lesson 11B	The sum of sine and cosine functions	Past papers: 21 marks · 30 min
Lesson 11C	Using the t-formula to solve equations	Past papers: 20 marks · 30 min
Review	Chapter 11 review exercise	Mixed practice

Each lesson runs in the same order: **concept → tutor demo → your attempt → worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions (about 30 minutes). Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

Trigonometric identities you need

From Year 11 Extension 1 (Chapter 17) and Advanced (trigonometric graphs). Everything in this chapter is built on these.

FORMULA SHEET · keep this page open while you work

COMPOUND-ANGLE FORMULAE

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta \qquad \sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$$

$$\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta \qquad \cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$$

$$\tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta} \qquad \tan(\alpha-\beta)=\frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta}$$

DOUBLE-ANGLE FORMULAE

$$\sin 2\theta=2\sin\theta\cos\theta \qquad \tan 2\theta=\frac{2\tan\theta}{1-\tan^2\theta}$$

$$\cos 2\theta=\cos^2\theta-\sin^2\theta=2\cos^2\theta-1=1-2\sin^2\theta$$

THE T-FORMULAE

Let $t=\tan\frac{1}{2}\theta$. Then

$$\sin\theta=\frac{2t}{1+t^2} \qquad \cos\theta=\frac{1-t^2}{1+t^2} \qquad \tan\theta=\frac{2t}{1-t^2}$$

PRODUCTS TO SUMS

$$2\sin A\cos B=\sin(A+B)+\sin(A-B) \qquad 2\cos A\sin B=\sin(A+B)-\sin(A-B)$$

$$2\cos A\cos B=\cos(A+B)+\cos(A-B) \qquad -2\sin A\sin B=\cos(A+B)-\cos(A-B)$$

WARM-UP · answers are in the answers booklet

1 Expand and simplify $\sin\left(x+\frac{\pi}{4}\right)$.

2 Find the exact value of $\cos 75°$ by writing $75°=45°+30°$.

3 Simplify $\frac{1-\cos 2\theta}{\sin 2\theta}$.

4 Solve $2\sin x = 1$ for $0 \leq x \leq 2\pi$.

5 Solve $\tan 2x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. (Start by writing down the restriction on $2x$.)

6 If $t = \tan \frac{1}{2}\theta$ and $t = \frac{1}{2}$, find the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

7 State the amplitude and period of $y = 2\sin\left(x - \frac{\pi}{4}\right)$, describe how it is obtained from $y = \sin x$, and sketch it for $0 \leq x \leq 2\pi$.

LESSON 11A

Equations involving compound angles

Use compound-angle and double-angle identities to turn an equation into one trigonometric function of one angle, then solve.

CONCEPT 1 · $\sin^2\theta$ and $\cos^2\theta$ in terms of $\cos 2\theta$

TWO NEW FORMS OF THE COS 2Θ FORMULA

Start from the second and third forms of $\cos 2\theta$ and make the square the subject.

$$\cos 2\theta = 2\cos^2\theta - 1 \qquad \cos 2\theta = 1 - 2\sin^2\theta$$
$$2\cos^2\theta = 1 + \cos 2\theta \qquad 2\sin^2\theta = 1 - \cos 2\theta$$
$$\cos^2\theta = \tfrac{1}{2} + \tfrac{1}{2}\cos 2\theta \qquad \sin^2\theta = \tfrac{1}{2} - \tfrac{1}{2}\cos 2\theta$$

Check: adding the two gives $\cos^2\theta + \sin^2\theta = 1$, as it must. These forms are used again in Chapter 12 to integrate $\sin^2 x$ and $\cos^2 x$.

WHY THIS HELPS

A squared sine or cosine becomes a **single cosine wave** $y = \frac{1}{2} \pm \frac{1}{2}\cos 2x$, so you can read off its amplitude, period and range.

TUTOR DEMO **Sketching a squared cosine**

Sketch $y = \cos^2 x$ for $0 \le x \le 2\pi$, and state its amplitude, period and range.

YOUR TURN **Same method**

Sketch $y = 2\cos^2 x$ for $0 \le x \le 2\pi$, and state its amplitude, period and range.

WORKED EXAMPLE **Keep for homework**

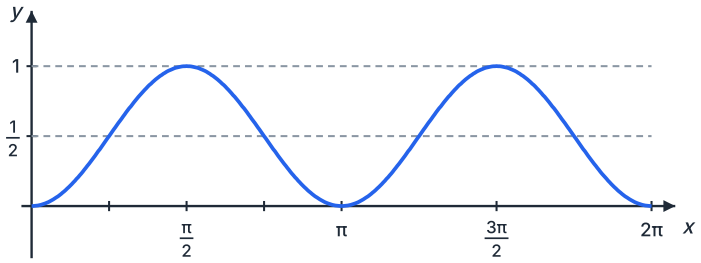
Sketch $y = \sin^2 x$, and state its amplitude, period and range.

SOLUTION

$y = \sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$

This is $y = \cos 2x$ turned upside down, stretched vertically by the factor $\frac{1}{2}$, then shifted up $\frac{1}{2}$.

Period π · **amplitude** $\frac{1}{2}$ · it oscillates about $y = \frac{1}{2}$, so the **range** is $0 \leq y \leq 1$.



CONCEPT 2 · more than one trig function, same angle

EQUATIONS WITH MORE THAN ONE TRIGONOMETRIC FUNCTION

- Use trigonometric identities (ratio and Pythagorean identities) to produce an equation in **only one** trigonometric function.
- Many trigonometric equations can be solved by more than one method.

THE DANGER OF SQUARING

Squaring both sides can introduce **extra solutions**. For example, $\sin x = 1$ for $0 \leq x \leq 2\pi$ has one solution $x = \frac{\pi}{2}$, but $\sin^2 x = 1$ has two solutions, $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$.

If you square, you **must check every answer** in the original equation.

TUTOR DEMO **Two methods**

Solve $4\tan\theta = 3\sec\theta$ for $0° \leq \theta \leq 360°$, correct to the nearest minute:

a using the ratio identities,

b by squaring both sides.

YOUR TURN **Ratio identities**

Solve $\tan\theta = 2\sin\theta$ for $0° \leq \theta \leq 360°$.

Careful: do not divide by $\sin\theta$. Move everything to one side and factor.

WORKED EXAMPLE **Keep for homework**

Solve $2\tan\theta = \sec\theta$ for $0° \leq \theta \leq 360°$: **a** using the ratio identities, **b** by squaring both sides.

SOLUTION

a

$$2\tan\theta = \sec\theta$$
$$\frac{2\sin\theta}{\cos\theta} = \frac{1}{\cos\theta}$$
$$\sin\theta = \tfrac{1}{2}$$
$$\theta = 30° \text{ or } 150°$$

b Squaring, $4\tan^2\theta = \sec^2\theta$

$$4\sec^2\theta - 4 = \sec^2\theta$$
$$\sec^2\theta = \tfrac{4}{3}$$
$$\cos\theta = \tfrac{\sqrt{3}}{2} \text{ or } -\tfrac{\sqrt{3}}{2}$$
$$\theta = 30°, 150°, 210° \text{ or } 330°$$

Checking each solution in the original equation, only $\theta = 30°$ or $150°$ work. (At $210°$, $\tan\theta > 0$ but $\sec\theta < 0$.)

CONCEPT 3 · equations involving different angles

DIFFERENT ANGLES IN ONE EQUATION

Use compound-angle identities to change all the trigonometric functions to functions of **one angle**.

GENERAL APPROACH TO ANY TRIGONOMETRIC EQUATION

1. First, try to get all the **angles** the same.
2. Then try to get all the **trigonometric functions** the same.

Then factor, solve each factor, and keep track of the domain of every angle that appears (if $0 \leq x \leq 2\pi$ then $0 \leq 2x \leq 4\pi$).

TUTOR DEMO **Double angle, then factor**

Solve $\sin 2x + \cos x = 0$ for $0 \leq x \leq 2\pi$.

YOUR TURN **Double angle, then a quadratic**

Solve $\cos 2x = \sin x$ for $0 \leq x \leq 2\pi$. (Choose the form of $\cos 2x$ that leaves only $\sin x$.)

TUTOR DEMO **Compound angle, then tan**

Use compound-angle formulae to show that $\cos\left(x - \frac{\pi}{3}\right) = \sin x$ reduces to $\tan x = 2 + \sqrt{3}$. Hence solve the equation for $0 \leq x \leq 2\pi$, given that $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$.

YOUR TURN **Same structure**

Solve $\sin\left(x+\frac{\pi}{6}\right)=\cos x$ for $0\le x\le 2\pi$.

WORKED EXAMPLE **Keep for homework**

Solve $\tan 4x=-\tan 2x$ for $0\le x\le\frac{\pi}{2}$, using the $\tan 2\theta$ formula.

SOLUTION

$$\tan 4x=-\tan 2x$$
$$\frac{2\tan 2x}{1-\tan^2 2x}=-\tan 2x$$
$$2\tan 2x=-\tan 2x+\tan^3 2x$$
$$\tan^3 2x-3\tan 2x=0$$
$$\tan 2x\left(\tan^2 2x-3\right)=0$$
$$\tan 2x=0,\quad \tan 2x=\sqrt{3}\quad\text{or}\quad \tan 2x=-\sqrt{3}$$

The restriction on $2x$ is $0\le 2x\le\pi$, so $2x=0$ or π, or $2x=\frac{\pi}{3}$, or $2x=\frac{2\pi}{3}$.

$$x=0,\ \tfrac{\pi}{6},\ \tfrac{\pi}{3}\text{ or }\tfrac{\pi}{2}$$

Alternatively, substitute $t=\tan 2x$ after the second line.

WORKED EXAMPLE **Two ways to make the angles the same**

Solve $\cos 2x=4\sin^2 x-14\cos^2 x$ for $0\le x\le 2\pi$: **a** by changing all the angles to x, **b** by changing all the angles to $2x$.

SOLUTION

a Using $\cos 2x=\cos^2 x-\sin^2 x$:

$$\cos^2 x-\sin^2 x=4\sin^2 x-14\cos^2 x$$
$$15\cos^2 x=5\sin^2 x$$
$$\tan^2 x=3$$
$$\tan x=\sqrt{3}\text{ or }-\sqrt{3}$$
$$x=\tfrac{\pi}{3},\tfrac{2\pi}{3},\tfrac{4\pi}{3}\text{ or }\tfrac{5\pi}{3}$$

b Using Concept 1:

$$\cos 2x=4\left(\tfrac{1}{2}-\tfrac{1}{2}\cos 2x\right)-14\left(\tfrac{1}{2}+\tfrac{1}{2}\cos 2x\right)$$
$$10\cos 2x=-5$$
$$\cos 2x=-\tfrac{1}{2}\quad(0\le 2x\le 4\pi)$$
$$2x=\tfrac{2\pi}{3},\tfrac{4\pi}{3},\tfrac{8\pi}{3}\text{ or }\tfrac{10\pi}{3}$$
$$x=\tfrac{\pi}{3},\tfrac{2\pi}{3},\tfrac{4\pi}{3}\text{ or }\tfrac{5\pi}{3}$$

CONCEPT 4 · homogeneous equations

HOMOGENEOUS EQUATIONS

- An equation is **homogeneous** in $\sin x$ and $\cos x$ if the sum of the indices of $\sin x$ and $\cos x$ in each term is the same.
- To solve it, **divide through by a suitable power of $\cos x$** to produce an equation in $\tan x$ alone.

The expansions of $\sin 2x$ and $\cos 2x$ are homogeneous of degree 2 in $\sin x$ and $\cos x$. Also, $1 = \sin^2 x + \cos^2 x$ can be regarded as homogeneous of degree 2.

Before dividing by $\cos^2 x$, check that $\cos x = 0$ is not a solution (substitute $x = \frac{\pi}{2}$), otherwise you would lose it.

TUTOR DEMO **Already homogeneous**

Solve $\sin^2 x - 2\sin x \cos x - 3\cos^2 x = 0$ for $0 \leq x \leq 2\pi$, giving answers correct to 3 decimal places where necessary.

YOUR TURN **Same method**

Solve $\sin^2 x + \sin x \cos x - 2\cos^2 x = 0$ for $0 \leq x \leq 2\pi$, giving answers correct to 3 decimal places where necessary.

WORKED EXAMPLE **Make it homogeneous first**

Solve $\sin 2x + \cos 2x = \sin^2 x + 1$ for $0 \le x \le 2\pi$.

SOLUTION

Expand, writing 1 as $\sin^2 x + \cos^2 x$ so that every term has degree 2:

$$2\sin x \cos x + (\cos^2 x - \sin^2 x) = \sin^2 x + (\sin^2 x + \cos^2 x)$$

$$3\sin^2 x - 2\sin x \cos x = 0$$

$$3\tan^2 x - 2\tan x = 0 \qquad (\div \cos^2 x)$$

$$\tan x\,(3\tan x - 2) = 0$$

$$\tan x = 0 \quad \text{or} \quad \tan x = \tfrac{2}{3}$$

Hence $x = 0,\ \pi$ or 2π, or $x \approx 0.588$ or 3.730. ($\cos x = 0$ is not a solution: at $x = \frac{\pi}{2}$, LHS $= -1$, RHS $= 2$.)

TEXTBOOK HOMEWORK · Exercise 11A

FOUNDATION

1 Consider the equation $\sin 2x - \cos x = 0$.

a By using a double-angle formula and then factoring, show that $\cos x = 0$ or $\sin x = \frac{1}{2}$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

2 Consider the equation $\cos 2x - \cos x = 0$.

a By using a double-angle formula and then factoring, show that $\cos x = 1$ or $-\frac{1}{2}$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

3 Consider the equation $\sin\left(x + \frac{\pi}{4}\right) = 2\cos\left(x - \frac{\pi}{4}\right)$.

a Use compound-angle formulae to show that $\tan x = -1$.

b Hence solve the equation for $0 \leq x \leq 2\pi$.

DEVELOPMENT

4 Use compound-angle formulae to solve, for $0 \leq \theta \leq 2\pi$:

a $\sin\left(\theta + \frac{\pi}{6}\right) = 2\sin\left(\theta - \frac{\pi}{6}\right)$

b $\cos 4\theta \cos\theta + \sin 4\theta \sin\theta = \frac{1}{2}$

c $\cos 3\theta = \cos 2\theta \cos\theta$ (write $\cos 3\theta$ as $\cos(2\theta + \theta)$)

5 Use double-angle formulae to solve, for $0 \leq x \leq 2\pi$:

a $\sin 2x + \sqrt{3}\cos x = 0$

b $3\sin x + \cos 2x = 2$

c $\tan 2x + \tan x = 0$

d $\sin 2x = \tan x$

6 Solve, for $0° \leq \theta \leq 360°$, giving solutions correct to the nearest minute where necessary:

a $2\sin 2\theta + \cos\theta = 0$

b $2\cos 2\theta + 4\cos\theta = 1$

c $3\cos 2\theta + \sin\theta = 1$

d $10\cos\theta + 13\cos\frac{1}{2}\theta = 5$

e $\cos^2 2\theta = \sin^2\theta$ (use $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta$)

f $\cos 2\theta + 3 = 3\sin 2\theta$ (write 3 as $3\cos^2\theta + 3\sin^2\theta$)

7 Consider the equation $\tan\left(\frac{\pi}{4} + \theta\right) = 3\tan\left(\frac{\pi}{4} - \theta\right)$.

a Show that $\tan^2\theta - 4\tan\theta + 1 = 0$.

b Hence use the quadratic formula to solve the equation for $0 \leq \theta \leq \pi$.

8 Given the equation $2\cos x - 1 = 2\cos 2x$:

a Show that $\cos x = \frac{1}{4}\left(1 + \sqrt{5}\right)$ or $\cos x = \frac{1}{4}\left(1 - \sqrt{5}\right)$.

b Hence solve the equation for $0 \leq x \leq 2\pi$. (Use $\cos \frac{\pi}{5} = \frac{1}{4}(1 + \sqrt{5})$ and $\cos \frac{3\pi}{5} = \frac{1}{4}(1 - \sqrt{5})$.)

9 a Show that $\sin 3x = 3\sin x - 4\sin^3 x$.

b Hence solve the equation $\sin 3x + \sin 2x = \sin x$, for $0 \leq x \leq 2\pi$.

10 Use compound-angle formulae, other trigonometric identities, and factoring to solve for $0° \leq \alpha \leq 360°$, giving solutions correct to the nearest minute where necessary.

a $\sec^2 \alpha = 2\sec \alpha$

b $\sec^2 \alpha - \tan \alpha - 3 = 0$

c $\sqrt{3}\operatorname{cosec}^2 \alpha = 4\cot \alpha$

d $\cot \alpha + 3\tan \alpha = 5\operatorname{cosec} \alpha$

ENRICHMENT

11 Consider the equation $\sin\theta + \cos\theta = \sin 2\theta$, for $0° \leq \theta \leq 360°$.

a By squaring both sides, show that $\sin^2 2\theta - \sin 2\theta - 1 = 0$.

b Hence solve for θ over the given domain, giving solutions correct to the nearest minute. (Beware: squaring can create invalid solutions.)

12 a Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

b By substituting $x = 2\cos\theta$, show that the equation $x^3 - 3x - 1 = 0$ has roots $2\cos 20°$, $-2\sin 10°$ and $-2\cos 40°$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **19 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(d)** 2 marks

Solve $3\sin 2x = \cos x$, for $0 \leq x \leq 2\pi$. Round your answers to 2 decimal places where necessary.

Hint: expand $\sin 2x$, then take out the common factor. Do not divide by $\cos x$.

PP 2 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(a)** 3 marks

Solve the equation $\sin 2x = \sqrt{2}\sin x$, for $0 \leq x \leq 2\pi$.

Hint: same idea as PP 1: factor out $\sin x$ and solve both factors.

PP 3 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q13(a)** 4 marks

(i) Show that $\tan 2x + \tan x = \dfrac{\sin 3x}{\cos 2x \cos x}$. 2

(ii) Hence solve the equation $\tan 2x + \tan x = 0$ for $0 \leq x \leq \dfrac{\pi}{2}$. 2

Hint: write both tangents as sine over cosine and add over a common denominator. The numerator is a compound-angle expansion.

PP 4 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(c)** 4 marks

(i) Show that $\sin 2\theta + \sin 4\theta - \sin 6\theta = 4 \sin 3\theta \sin 2\theta \sin \theta$. 2

(ii) Hence, solve $\sin 2\theta + \sin 4\theta = \sin 6\theta$ for $0 \leq \theta \leq \pi$. 2

Hint: write $\sin 4\theta = \sin(3\theta + \theta)$, $\sin 2\theta = \sin(3\theta - \theta)$ and $\sin 6\theta = 2 \sin 3\theta \cos 3\theta$; then $\cos 3\theta$ and $\cos \theta$ are $\cos(2\theta \pm \theta)$.

PP 5 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)** 6 marks

(i) Show $\sin(x - y) \sin(x + y) = \sin^2 x - \sin^2 y$. 3

(ii) Solve $\sin^2 3x - \sin^2 x = \sin 4x$ for $0 \leq x \leq \pi$. 3

Hint: in (ii) use part (i) with $3x$ and x in place of x and y, then factor out $\sin 4x$.

LESSON 11B

The sum of sine and cosine functions

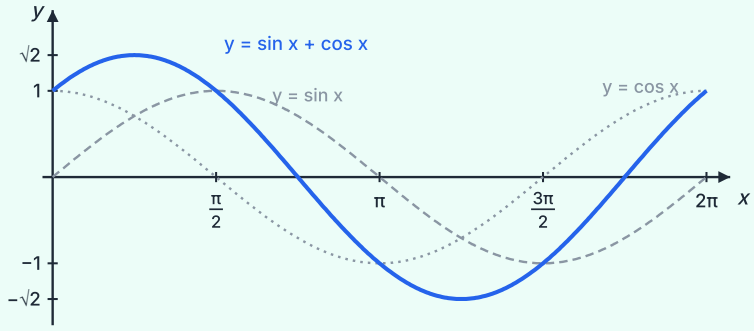
Write $a\sin x + b\cos x$ as a single wave $R\sin(x \pm \alpha)$ or $R\cos(x \pm \alpha)$, then use it to sketch, find maximum and minimum values, and solve equations and inequations.

CONCEPT 1 · adding a sine wave and a cosine wave

SKETCHING Y = SIN X + COS X BY GRAPHICAL METHODS

Add the heights of $y = \sin x$ and $y = \cos x$ at convenient points.

- The new graph has the same period 2π. It looks like a wave, and within $[0, 2\pi]$ its zeroes are at $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$, where $\sin x$ and $\cos x$ take opposite values.
- The amplitude is bigger than 1. At $x = \frac{\pi}{4}$ the value is $\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2} = \sqrt{2}$, so if the maximum occurs there, the amplitude is $\sqrt{2}$.



This suggests $y = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$: the sine curve stretched by $\sqrt{2}$ and shifted left $\frac{\pi}{4}$. Check by expanding:

$$\sqrt{2}\sin\left(x + \tfrac{\pi}{4}\right) = \sqrt{2}\left(\sin x \cos\tfrac{\pi}{4} + \cos x \sin\tfrac{\pi}{4}\right) = \sin x + \cos x, \quad \text{because } \cos\tfrac{\pi}{4} = \sin\tfrac{\pi}{4} = \frac{1}{\sqrt{2}}.$$

CONCEPT 2 · the auxiliary-angle method

AUXILIARY-ANGLE METHOD

- Any function $f(x) = a\sin x + b\cos x$ (a, b constants, not both zero) can be written in any one of the four forms

  $y = R\sin(x - \alpha)$ $\quad y = R\sin(x + \alpha)$ $\quad y = R\cos(x - \alpha)$ $\quad y = R\cos(x + \alpha)$

  where $R > 0$ and $0° \le \alpha < 360°$. The constant $R = \sqrt{a^2 + b^2}$ is the same for all forms, but the **auxiliary angle** α depends on which form is chosen.
- To begin, **expand the standard form and equate coefficients** of $\sin x$ and $\cos x$. Be careful to identify the **quadrant** in which α lies.

THE ALGORITHM

1. Expand the required form with the compound-angle formula.
2. Equate coefficients of $\sin x$ and $\cos x$: this gives $R\cos\alpha = \ldots$ (1) and $R\sin\alpha = \ldots$ (2).
3. Square and add: $R^2 = a^2 + b^2$, and $R > 0$.
4. Use the **signs** of $\cos\alpha$ and $\sin\alpha$ to find the quadrant, then the related angle.

Maximum value $= R$, minimum value $= -R$. The graph is a sine or cosine wave with amplitude R, shifted sideways by α.

TUTOR DEMO **Exact auxiliary angle, two forms**

Express $y = \sqrt{3}\sin x + \cos x$ in the two forms **a** $R\sin(x+\alpha)$, **b** $R\cos(x+\alpha)$, where $R > 0$ and $0 \le \alpha < 2\pi$. Then sketch the curve for $0 \le x \le 2\pi$, showing all intercepts and turning points.

YOUR TURN **One form**

Express $\sin x - \cos x$ in the form $R\sin(x-\alpha)$, where $R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$. State the maximum and minimum values of $\sin x - \cos x$, and sketch $y = \sin x - \cos x$ for $0 \le x \le 2\pi$.

WORKED EXAMPLE 1 **Keep for homework**

Express $y = \sin x + \cos x$ in the two forms **a** $R\sin(x+\alpha)$, **b** $R\cos(x+\alpha)$, where $R > 0$ and $0 \le \alpha < 2\pi$. Then sketch the curve, showing all intercepts and turning points in $0 \le x \le 2\pi$.

SOLUTION

a Expanding,
$R\sin(x+\alpha) = R\sin x\cos\alpha + R\cos x\sin\alpha$,
so for all x,

$$\sin x + \cos x = R\sin x\cos\alpha + R\cos x\sin\alpha.$$

Equating coefficients:

$$\sin x: \quad R\cos\alpha = 1 \quad (1)$$
$$\cos x: \quad R\sin\alpha = 1 \quad (2)$$

Squaring and adding, $R^2 = 2$, and because $R > 0$, $R = \sqrt{2}$.

$$\cos\alpha = \frac{1}{\sqrt{2}} > 0, \qquad \sin\alpha = \frac{1}{\sqrt{2}} > 0$$

so α is in the 1st quadrant, with related angle $\frac{\pi}{4}$:

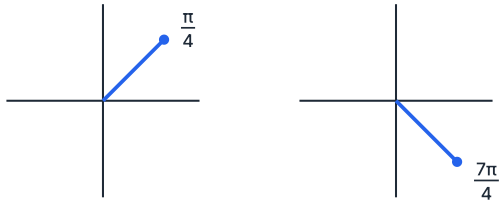
$$\sin x + \cos x = \sqrt{2}\sin\left(x + \tfrac{\pi}{4}\right)$$

b Expanding,
$R\cos(x+\alpha) = R\cos x\cos\alpha - R\sin x\sin\alpha$.

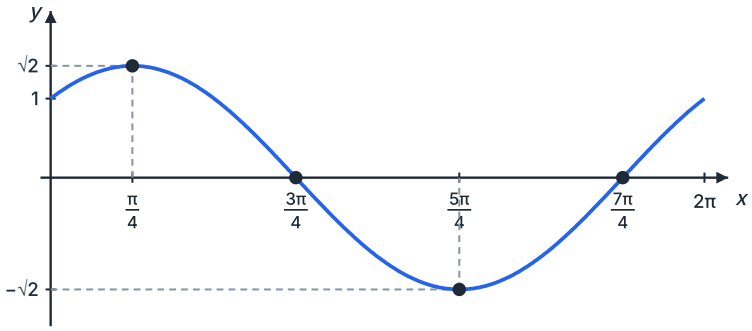
$$\cos x: \quad R\cos\alpha = 1 \quad (1)$$
$$\sin x: \quad R\sin\alpha = -1 \quad (2)$$

Squaring and adding, $R = \sqrt{2}$. Now $\cos\alpha > 0$ and $\sin\alpha < 0$, so α is in the 4th quadrant, with related angle $\frac{\pi}{4}$:

$$\sin x + \cos x = \sqrt{2}\cos\left(x + \tfrac{7\pi}{4}\right)$$



The graph is $y = \sin x$ shifted left $\frac{\pi}{4}$ and stretched vertically by $\sqrt{2}$. The x-intercepts are $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$; there is a maximum of $\sqrt{2}$ when $x = \frac{\pi}{4}$ and a minimum of $-\sqrt{2}$ when $x = \frac{5\pi}{4}$.



CONCEPT 3 · approximating the auxiliary angle

WHEN A IS NOT A SPECIAL ANGLE

Unless special angles are involved, approximate the auxiliary angle on the calculator. Degrees or radians may be used.

Calculator note. In Worked Example 2 below the exact value is $\alpha = 180° - \sin^{-1}\frac{3}{5}$, because α is in the second quadrant. Store this value in the calculator's memory and use it whenever α is needed later. Re-entering a rounded value may lead to rounding errors.

TUTOR DEMO **Approximate α**

Express $5\sin x + 12\cos x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0° \leq \alpha < 360°$, giving α correct to the nearest minute.

YOUR TURN **Same method**

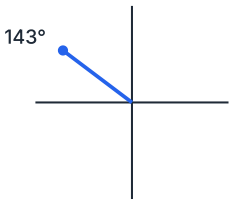
Express $4\sin x + 3\cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0° \leq \alpha < 360°$, giving α correct to the nearest minute.

WORKED EXAMPLE 2 **Keep for homework**

a Express $y = 3\sin x - 4\cos x$ in the form $y = R\cos(x - \alpha)$, where $R > 0$ and $0° \leq \alpha < 360°$, giving α correct to the nearest degree.
b Sketch the curve, showing all intercepts and turning points in $-180° \leq x \leq 180°$, correct to the nearest degree.

SOLUTION

a Expanding,
$R\cos(x - \alpha) = R\cos x\cos\alpha + R\sin x\sin\alpha$, so for all x,



$$3\sin x - 4\cos x = R\cos x\cos\alpha + R\sin x\sin\alpha.$$
$$\cos x: \quad R\cos\alpha = -4 \quad (1)$$
$$\sin x: \quad R\sin\alpha = 3 \quad (2)$$

Squaring and adding, $R^2 = 25$, so $R = 5$.

$$\cos\alpha = -\tfrac{4}{5}, \qquad \sin\alpha = \tfrac{3}{5}$$

so α is in the 2nd quadrant, with related angle about $37°$. Hence

$$3\sin x - 4\cos x = 5\cos(x - \alpha), \text{ where } \alpha \approx 143°.$$

b The graph is $y = \cos x$ shifted right by $\alpha \approx 143°$ and stretched vertically by 5. The x-intercepts are $x \approx 53°$ and $x \approx -127°$; there is a maximum of 5 when $x \approx 143°$ and a minimum of -5 when $x \approx -37°$. (The y-intercept is -4.)

CONCEPT 4 · solving a sin x + b cos x = c, and inequations

AUXILIARY-ANGLE METHOD FOR A SIN X + B COS X = C

- Get the LHS into one of the four forms $R\sin(x + \alpha)$, $R\sin(x - \alpha)$, $R\cos(x + \alpha)$ or $R\cos(x - \alpha)$.
- Then solve the resulting equation. Always keep track of the **restriction on the compound angle**: if $0° \leq x \leq 360°$, then $-\alpha \leq x - \alpha \leq 360° - \alpha$.
- For an **inequation**, solve the equation first, then use the graph of the wave and the horizontal line to read off where the curve is above or below the line.

TUTOR DEMO **Solve with the auxiliary angle**

Using your answer to the tutor demo in Concept 3, solve $5\sin x + 12\cos x = 6.5$ for $0° \leq x \leq 360°$, correct to the nearest minute.

YOUR TURN **Same method**

Using your answer to the Concept 3 attempt, solve $4\sin x + 3\cos x = 2.5$ for $0° \leq x \leq 360°$, correct to the nearest minute.

TUTOR DEMO **An inequation**

Using $\sqrt{3}\sin x + \cos x = 2\sin\left(x + \frac{\pi}{6}\right)$, solve $\sqrt{3}\sin x + \cos x \geq 1$ for $0 \leq x \leq 2\pi$.

YOUR TURN **Same method**

Using $\sin x - \cos x = \sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$, solve $\sin x - \cos x > 1$ for $0 \leq x \leq 2\pi$.

WORKED EXAMPLE 3 **Keep for homework**

a Using Worked Example 2, solve $3\sin x - 4\cos x = -2$ for $-180° \leq x \leq 180°$, correct to the nearest degree.
b Hence use the graph to solve $3\sin x - 4\cos x \leq -2$ for $-180° \leq x \leq 180°$.

SOLUTION

a Using $3\sin x - 4\cos x = 5\cos(x - \alpha)$, where $\alpha \approx 143°$:

$$5\cos(x - \alpha) = -2, \quad \text{where } -323° \leq x - \alpha \leq 37°$$
$$\cos(x - \alpha) = -\frac{2}{5}$$

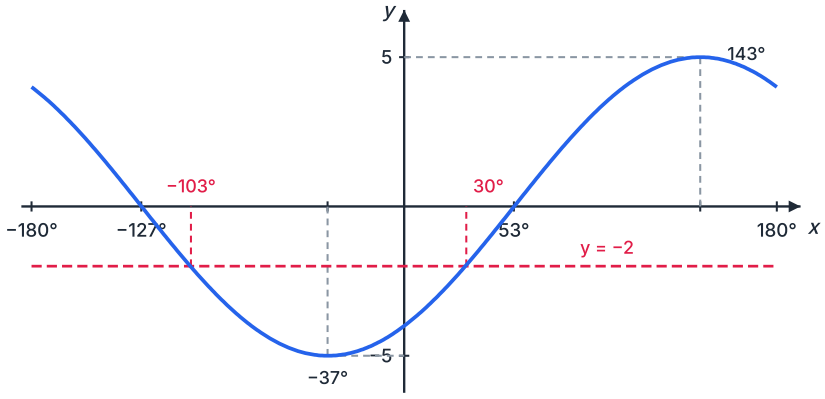
Hence $x - \alpha$ is in quadrant 2 or 3, with related angle about $66°$, so

$$x - \alpha \approx -114° \text{ or } -246°$$
$$x \approx 30° \text{ or } -103°$$

Be careful to use the calculator's memory here. Never re-enter approximations of the angles.

b The graph of $y = 3\sin x - 4\cos x$ with the line $y = -2$ added verifies the two answers in part **a**, and shows that the solution of the inequation is

$$-103° \leq x \leq 30°.$$



TEXTBOOK HOMEWORK · Exercise 11B

FOUNDATION

1 Find R and α exactly, if $R > 0$ and $0 \leq \alpha < 2\pi$, and:

a $R\sin\alpha = \sqrt{3}$ and $R\cos\alpha = 1$

b $R\sin\alpha = 3$ and $R\cos\alpha = 3$

2 Find R (exactly) and α (correct to the nearest minute), if $R > 0$ and $0° \leq \alpha < 360°$, and:

a $R\sin\alpha = 5$ and $R\cos\alpha = 12$

b $R\cos\alpha = 2$ and $R\sin\alpha = 4$

3 a If $\cos x - \sin x = A\cos(x + \alpha)$, show that $A\cos\alpha = 1$ and $A\sin\alpha = 1$.

b Find the positive value of A by squaring and adding.

c Find α, if $0 \leq \alpha < 2\pi$.

d State the maximum and minimum values of $\cos x - \sin x$, and the first positive values of x for which they occur.

e Solve the equation $\cos x - \sin x = -1$, for $0 \leq x \leq 2\pi$.

f Write down the amplitude and period of $\cos x - \sin x$. Hence sketch $y = \cos x - \sin x$, for $0 \leq x \leq 2\pi$. Indicate on your sketch the line $y = -1$ and the solutions of the equation in part e.

4 a If $\sqrt{3}\cos x - \sin x = B\cos(x+\theta)$, show that $B\cos\theta = \sqrt{3}$ and $B\sin\theta = 1$.

b Find B, if $B > 0$, by squaring and adding.

c Find θ, if $0 \leq \theta < 2\pi$.

d State the greatest and least possible values of $\sqrt{3}\cos x - \sin x$, and the values of x closest to $x = 0$ for which they occur.

e Solve the equation $\sqrt{3}\cos x - \sin x = 1$, for $0 \leq x \leq 2\pi$.

f Sketch $y = \sqrt{3}\cos x - \sin x$, for $0 \leq x \leq 2\pi$. On the same diagram, sketch the line $y = 1$. Indicate the solutions of the equation in part e.

5 Let $4\sin x - 3\cos x = A\sin(x-\alpha)$, where $A > 0$ and $0° \leq \alpha < 360°$.

a Show that $A\cos\alpha = 4$ and $A\sin\alpha = 3$.

b Show that $A = 5$ and $\alpha = \tan^{-1}\frac{3}{4}$.

c Hence solve the equation $4\sin x - 3\cos x = 5$, for $0° \leq x \leq 360°$. Give the solution(s) correct to the nearest minute.

6 Let $\cos x - 3\sin x = D\cos(x+\phi)$, where $D > 0$ and $0° \leq \phi < 360°$.

a Show that $D = \sqrt{10}$ and $\phi = \tan^{-1}3$.

b Hence solve $\cos x - 3\sin x = 3$, for $0° \leq x \leq 360°$. Give the solutions correct to the nearest minute where necessary.

7 Consider the equation $\sqrt{5}\sin x + 2\cos x = -2$.

a Transform the LHS into the form $C\sin(x+\alpha)$, where $C > 0$ and $0° \leq \alpha < 360°$.

b Find, correct to the nearest minute where necessary, the solutions of the equation, for $0° \leq x \leq 360°$.

DEVELOPMENT

8 Solve each equation, for $0° \leq x \leq 360°$, by transforming the LHS into a single-term sine or cosine function. Give solutions correct to the nearest minute.

a $3\sin x + 5\cos x = 4$

b $6\sin x - 5\cos x = 7$

c $7\cos x - 2\sin x = 5$

d $9\cos x + 7\sin x = 3$

9 a (i) Express $\sqrt{3}\cos x + \sin x$ in the form $A\cos(x+\theta)$, where $A > 0$ and $0 < \theta < 2\pi$. (ii) Hence solve $\sqrt{3}\cos x + \sin x = 1$, for $0 \leq x < 2\pi$.

b (i) Express $\cos x - \sin x$ in the form $B\sin(x+\alpha)$, where $B > 0$ and $0 < \alpha < 2\pi$. (ii) Hence solve $\cos x - \sin x = 1$, for $0 \leq x < 2\pi$.

c (i) Express $\sin x - \sqrt{3}\cos x$ in the form $C\sin(x+\beta)$, where $C > 0$ and $0 < \beta < 2\pi$. (ii) Hence solve $\sin x - \sqrt{3}\cos x = -1$, for $0 \leq x < 2\pi$.

d (i) Express $-\cos x - \sin x$ in the form $D\cos(x-\phi)$, where $D > 0$ and $0 < \phi < 2\pi$. (ii) Hence solve $-\cos x - \sin x = 1$, for $0 \leq x < 2\pi$.

10 a (i) Express $2\cos x - \sin x$ in the form $R\sin(x+\alpha)$, where $R > 0$ and $0° < \alpha < 360°$. (Write α correct to the nearest minute.) (ii) Hence solve $2\cos x - \sin x = 1$, for $0° \leq x < 360°$. Give the solutions correct to the nearest minute where necessary.

b (i) Express $-3\sin x - 4\cos x$ in the form $S\cos(x-\beta)$, where $S > 0$ and $0 < \beta < 2\pi$. (Write β correct to four decimal places.) (ii) Hence solve $-3\sin x - 4\cos x = 2$, for $0 \leq x < 2\pi$. Give the solutions correct to two decimal places.

11 Solve, for $0° \leq x \leq 360°$, giving solutions correct to the nearest minute:

a $2\sec x - 2\tan x = 5$ b $2\operatorname{cosec} x + 5\cot x = 3$

12 a Given the equation $\sin\theta + \cos\theta = \cos 2\theta$, show that $\tan\theta = -1$ or $\cos\theta - \sin\theta = 1$.

b Hence solve $\sin\theta + \cos\theta = \cos 2\theta$, for $0 \leq \theta < 2\pi$.

13 Solve, for $0 \leq x \leq 2\pi$:

a $\sin x - \cos x = \sqrt{1.5}$ b $\sqrt{3}\sin 2x - \cos 2x = 2$ c $\sin 4x + \cos 4x = 1$

14 a (i) Show that $\sin x - \cos x = \sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$. (ii) Hence sketch the graph of $y = \sin x - \cos x$, for $0 \leq x \leq 2\pi$. (iii) Use your sketch to determine the values of x in the domain $0 \leq x \leq 2\pi$ for which $\sin x - \cos x > 1$.

b Use a similar approach to part a to solve, for $0 \leq x \leq 2\pi$:
(i) $\sin x + \sqrt{3}\cos x \leq 1$ (ii) $\sin x - \sqrt{3}\cos x < -1$ (iii) $\left|\sqrt{3}\sin x + \cos x\right| < 1$ (iv) $\cos x - \sin x \geq \frac{1}{2}\sqrt{2}$

ENRICHMENT

15 a Show that if $\cos(x - \alpha) = \cos\beta$, then $\tan x = \tan(\alpha + \beta)$ or $\tan x = \tan(\alpha - \beta)$.

b Show that $2\cos x + 11\sin x = 5\sqrt{5}\cos\left(x - \tan^{-1}\frac{11}{2}\right)$.

c Consider the equation $2\cos x + 11\sin x = 10$, for $0 \leq x < 2\pi$. (i) By writing the equation in the form $\cos(x - \alpha) = \cos\beta$ and using part a, show that $\tan x = \frac{4}{3}$ or $-\frac{24}{7}$. (ii) Deduce that the equation has roots $\tan^{-1}\frac{4}{3}$ and $\pi - \tan^{-1}\frac{24}{7}$. (iii) Prove that one of the roots is twice the other.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q9** 1 mark

The expression $3\cos x + 2\sin x$ is written in the form $R\cos(x-\theta)$, where $R > 0$ and $0 \leq \theta \leq \dfrac{\pi}{2}$.

What is the value of $\tan\theta$?

(A) $-\dfrac{3}{2}$

(B) $-\dfrac{2}{3}$

(C) $\dfrac{2}{3}$

(D) $\dfrac{3}{2}$

Hint: expand $R\cos(x-\theta)$ and equate coefficients, then divide one equation by the other.

PP 2 **Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(e)** 4 marks

(i) Express $3\cos x + 2\sin x$ in the form $R\cos(x-\alpha)$, where $R > 0$ and $0° < \alpha < 90°$. 2

(ii) Hence, or otherwise, solve $3\cos x + 2\sin x = \sqrt{13}$, for $0° \leq x \leq 360°$. Give your answer in degrees correct to two decimal places. 2

Hint: $\sqrt{13}$ is the value of R. What must $\cos(x-\alpha)$ equal?

PP 3 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)** 4 marks

(i) Express $6\cos x + 8\sin x$ in the form $A\cos(x-\alpha)$, where $A > 0$ and $0 \leq \alpha \leq \dfrac{\pi}{2}$. 2

(ii) Hence, or otherwise, solve the equation $6\cos x + 8\sin x = 5$ for $0 \leq x \leq 2\pi$. Write your answers correct to three decimal places. 2

Hint: keep α in the calculator's memory, and write down the range of $x - \alpha$ before solving.

PP 4 **North Sydney Boys High School, 2023, Year 12 Mathematics Extension 1 Assessment Task 3, Q11(b)** 7 marks

(i) Express $\sin 2x + \sqrt{3}\cos 2x$ in the form of $R\sin(2x+\alpha)$. 2

(ii) Find the greatest value of $\sin 2x + \sqrt{3}\cos 2x$ and the values of x, $0 \leq x \leq 2\pi$, at which it occurs. 2

(iii) Hence, or otherwise, solve $\sin 2x + \sqrt{3}\cos 2x = 1$, $0 \leq x \leq 2\pi$. 3

Hint: the method is the same with $2x$ in place of x. If $0 \leq x \leq 2\pi$, what is the range of $2x + \alpha$? Expect four answers in (iii).

PP 5 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(c)** 5 marks

A machine produces a sound which can be modelled using the equation $y = \cos t - \sqrt{3}\sin t$, where y gives a measure of the sound at time t seconds after the machine is switched on. The sound is audible to humans when $y \geq 1$.

(i) Express $\cos t - \sqrt{3}\sin t$ in the form $R\cos(t+\alpha)$, where $R > 0$ and $0 \leq \alpha < \dfrac{\pi}{2}$. 2

(ii) Hence find the time intervals during which the noise is audible to humans for $0 < t \leq 4\pi$. 3

Hint: solve the equation $R\cos(t+\alpha) = 1$ first, then sketch the wave and the line $y = 1$ to see where the curve is above the line.

LESSON 11C

Using the t-formula to solve equations

Substitute $t = \tan \frac{1}{2}x$ to turn $a \sin x + b \cos x = c$ into a quadratic in t, and never lose the solution $x = 180°$.

CONCEPT 1 · the t-formula substitution

A DIFFERENT METHOD FOR A SIN X + B COS X = C

Substitute $t = \tan \frac{1}{2}x$, using

$$\sin x = \frac{2t}{1+t^2} \quad \text{and} \quad \cos x = \frac{1-t^2}{1+t^2},$$

then multiply through by $1 + t^2$. Everything quickly reduces to a **quadratic equation in t**.

Advantages: the method is far more automatic, and only a single approximation is involved.

Two disadvantages:

- The intuition about the LHS being a shifted wave function is lost.
- If $x = 180°$ happens to be a solution, it will not be found by this method, because $\tan \frac{1}{2}x$ is not defined at $x = 180°$.

Remember the restriction on $\frac{1}{2}x$: if $-180° \le x \le 180°$ then $-90° \le \frac{1}{2}x \le 90°$; if $0° \le x \le 360°$ then $0° \le \frac{1}{2}x \le 180°$.

TUTOR DEMO **t-formula, quadratic in t**

Solve $5 \sin x + 2 \cos x = 3$, for $-180° \le x \le 180°$, correct to the nearest minute, using the substitution $t = \tan \frac{1}{2}x$.

YOUR TURN **Same method**

Solve $3\sin x + 4\cos x = 1$, for $-180° \leq x \leq 180°$, correct to the nearest minute, using the substitution $t = \tan \frac{1}{2}x$.

WORKED EXAMPLE 1 **Keep for homework**

Solve $3\sin x - 4\cos x = -2$, for $-180° \leq x \leq 180°$, correct to the nearest minute, using the substitution $t = \tan \frac{1}{2}x$.

SOLUTION

Using $\sin x = \dfrac{2t}{1+t^2}$ and $\cos x = \dfrac{1-t^2}{1+t^2}$, the equation becomes

$$\frac{6t}{1+t^2} - \frac{4-4t^2}{1+t^2} = -2, \quad \text{provided that } x \neq 180°$$

$$6t - 4 + 4t^2 = -2 - 2t^2$$

$$6t^2 + 6t - 2 = 0$$

$$3t^2 + 3t - 1 = 0, \quad \text{which has discriminant } \Delta = 21$$

$$\tan \tfrac{1}{2}x = -\tfrac{1}{2} + \tfrac{1}{6}\sqrt{21} \text{ or } -\tfrac{1}{2} - \tfrac{1}{6}\sqrt{21}$$

Because $-180° \leq x \leq 180°$, the restriction on $\frac{1}{2}x$ is $-90° \leq \frac{1}{2}x \leq 90°$, so

$$\tfrac{1}{2}x = 14.775\,961\ldots° \text{ or } -51.645\,859\ldots° \quad \text{(only round at the last step)}$$

$$x \approx 29°33' \text{ or } -103°18'$$

Check $x = \pm 180°$, which the substitution cannot reach: LHS $= 3 \times 0 - 4 \times (-1) = 4 \neq -2$, so neither is a solution.

Compare with Worked Example 3 of Lesson 11B, which solved the same equation to the nearest degree.

CONCEPT 2 · the problem when x = 180° is a solution

ALWAYS BE READY TO ADD X = 180°

The substitution $t = \tan \frac{1}{2}x$ fails when $x = 180°$, because $\tan 90°$ is undefined. We must always be aware of this possibility, and be prepared to add this answer to the final solution. The situation can easily be recognised in either of these ways:

- The terms in t^2 **cancel out**, leaving a linear equation in t.
- The coefficient of $\cos x$ is the **opposite** of the constant term.

Check by substituting $x = 180°$ (where $\sin x = 0$, $\cos x = -1$) into the original equation.

TUTOR DEMO **The t^2 terms cancel**

Solve $5\sin x - 3\cos x = 3$, for $0° \leq x \leq 360°$, by using the substitution $t = \tan \frac{1}{2}x$. Give answers correct to the nearest minute where necessary.

YOUR TURN **Same check**

Solve $2\sin x - \cos x = 1$, for $0° \leq x \leq 360°$, by using the substitution $t = \tan \frac{1}{2}x$. Give answers correct to the nearest minute where necessary.

WORKED EXAMPLE 2 **Keep for homework**

Solve $7\sin x - 4\cos x = 4$, for $0° \leq x \leq 360°$, by using the substitution $t = \tan \frac{1}{2}x$.

SOLUTION

Substituting $t = \tan \frac{1}{2}x$ gives

$$\frac{14t}{1+t^2} - \frac{4-4t^2}{1+t^2} = 4, \quad \text{provided that } x \neq 180°$$

$$14t - 4 + 4t^2 = 4 + 4t^2$$

$$14t = 8$$

Warning: the terms in t^2 have cancelled out, so check $x = 180°$!

$$\tan \tfrac{1}{2}x = \tfrac{4}{7}, \qquad x \approx 59°29'$$

But $x = 180°$ is also a solution, because then LHS $= 7 \times 0 - 4 \times (-1) = 4 =$ RHS. So $x = 180°$ or $x \approx 59°29'$.

CONCEPT 3 · choosing a method

SOLVING EQUATIONS OF THE FORM A SIN X + B COS X = C

- **The auxiliary-angle method:** get the LHS into one of the forms $R\sin(x+\alpha)$, $R\sin(x-\alpha)$, $R\cos(x+\alpha)$ or $R\cos(x-\alpha)$, then solve the resulting equation.
- **Using the t-formulae:** substitute $t = \tan\frac{1}{2}x$ and then solve the resulting quadratic in t. Be aware that $x = 180°$ may also be a solution if
  - the terms in t^2 cancel out, leaving a linear equation in t, or equivalently,
  - the coefficient of $\cos x$ is the opposite of the constant term.

DOUBLE ANGLES: USE T = TAN X

For an equation in $\sin 2x$ and $\cos 2x$, use the substitution $t = \tan x$, so that $\sin 2x = \dfrac{2t}{1+t^2}$ and $\cos 2x = \dfrac{1-t^2}{1+t^2}$. Now the value that can be lost is $2x = 180°$ (and $540°$), that is, $x = 90°$ and $x = 270°$.

WORKED EXAMPLE 3 **t = tan x for a double angle**

Solve $\sin 2x + 3\cos 2x = 3$, for $0 \le x \le 2\pi$, using the substitution $t = \tan x$. Give answers correct to 3 decimal places where necessary.

SOLUTION

$$\frac{2t}{1+t^2} + \frac{3(1-t^2)}{1+t^2} = 3, \quad \text{provided that } x \neq \tfrac{\pi}{2}, \tfrac{3\pi}{2}$$

$$2t + 3 - 3t^2 = 3 + 3t^2$$

$$6t^2 - 2t = 0$$

$$2t(3t-1) = 0$$

$$\tan x = 0 \quad \text{or} \quad \tan x = \tfrac{1}{3}$$

So $x = 0,\ \pi,\ 2\pi$, or $x \approx 0.322$ or $x \approx 0.322 + \pi \approx 3.463$.

Check $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$: $\sin\pi + 3\cos\pi = -3 \neq 3$, so neither is a solution.

Solutions: $x = 0,\ 0.322,\ \pi,\ 3.463,\ 2\pi$.

TEXTBOOK HOMEWORK · Exercise 11C

FOUNDATION

1 Consider the equation $\cos x - \sin x = 1$, where $0 \leq x \leq 2\pi$.

a Using the substitutions $\sin x = \dfrac{2t}{1+t^2}$ and $\cos x = \dfrac{1-t^2}{1+t^2}$, where $t = \tan \frac{1}{2}x$, show that the equation can be written as $t^2 + t = 0$.

b Hence show that $\tan \frac{1}{2}x = 0$ or -1, where $0 \leq \frac{1}{2}x \leq \pi$.

c Hence solve the given equation for x.

2 Consider the equation $\sqrt{3}\sin x + \cos x = 1$.

a Show that the equation can be written as $t^2 = \sqrt{3}\,t$, where $t = \tan \frac{1}{2}x$.

b Hence solve the equation, for $0 \leq x \leq 2\pi$.

3 a Show that the equation $4\cos x + \sin x = 1$ can be written as $(5t+3)(t-1) = 0$, where $t = \tan \frac{1}{2}x$.

b Hence solve the equation, for $0° \leq x \leq 360°$. Give the solutions correct to the nearest minute where necessary.

4 a Show that the equation $3\sin x - 2\cos x = 2$ can be written as $3t - 2 = 0$, where $t = \tan \frac{1}{2}x$.

b Hence solve the equation for $0° \leq x \leq 360°$, giving solutions correct to the nearest minute where necessary. (Remember to check $x = 180°$ as a possible solution, given that the resulting equation in t is linear.)

5 a Show that the equation $6\sin x - 4\cos x = 5$ can be written as $t^2 - 12t + 9 = 0$, where $t = \tan \frac{1}{2}x$.

b Show that $\tan \frac{1}{2}x = 6 + 3\sqrt{3}$ or $6 - 3\sqrt{3}$.

c Hence show that $77°35'$ and $169°48'$ are the solutions (correct to the nearest minute) of the given equation over the domain $0° \leq x \leq 360°$.

DEVELOPMENT

6 Solve each equation, for $0° \leq x \leq 360°$, using the t-formulae, where $t = \tan \frac{1}{2}x$. Give solutions correct to the nearest minute where necessary.

a $5\sin x + 4\cos x = 5$

b $7\cos x - 6\sin x = 2$

c $3\sin x - 2\cos x = 1$

d $5\cos x + 6\sin x = -5$

7 Use t-formulae to prove that the equation $8\tan\theta - 4\sec\theta = 1$ has solutions $2\tan^{-1}\frac{1}{3}$ and $2\tan^{-1}5$ in the domain $0 \leq \theta \leq 2\pi$.

8 Solve the equation $2\sin 2x + \cos 2x = 2$, for $0° \leq x \leq 360°$, using the substitution $t = \tan x$. Give solutions correct to the nearest tenth of a degree where necessary.

9 Suppose that $a\cos x = 1 + \sin x$, where $0° < x < 90°$.

a Prove that $\dfrac{a-1}{a+1} = t$, where $t = \tan\frac{1}{2}x$.

b Hence find, correct to the nearest minute, the acute angle x that satisfies the equation $2\cos x - \sin x = 1$.

10 Let θ_1 and θ_2, where $\theta_1 > \theta_2$, be the solutions of the equation $6\cos\theta + 17\sin\theta = 18$ in the domain $0 \leq \theta \leq 2\pi$. Show that $\tan\left(\dfrac{\theta_1 - \theta_2}{2}\right) = \dfrac{1}{18}$.

ENRICHMENT

11 Consider the equation $a\cos x + b\sin x = c$, where a, b and c are constants.

a Show that the equation can be written in the form $(a+c)t^2 - 2bt - (a-c) = 0$, where $t = \tan\frac{1}{2}x$.

b Show that the root(s) of the equation are real if $c^2 \leq a^2 + b^2$.

c Suppose that $\tan\frac{1}{2}\alpha$ and $\tan\frac{1}{2}\beta$ are distinct real roots of the quadratic equation in part a. Prove that $\tan\frac{1}{2}(\alpha+\beta) = \dfrac{b}{a}$.

12 Given the equation $(2k-1)\cos\theta + (k+2)\sin\theta = 2k+1$, where k is a constant, use t-formulae to prove that $\tan\theta = \dfrac{4}{3}$ or $\dfrac{2k}{k^2-1}$.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **20 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3** 1 mark

Given that $t = \tan\dfrac{\theta}{2}$, then $\tan\theta\sin\theta$ is equal to

A. $\dfrac{4t^2}{1-t^4}$ B. $\dfrac{2t^2}{1-t^4}$ C. $\dfrac{4t^2}{1+t^4}$ D. $\dfrac{2t^2}{1+t^2}$

Hint: multiply the t-formulae for $\tan\theta$ and $\sin\theta$, and use the difference of two squares in the denominator.

PP 2 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)** 3 marks

Using $t = \tan\dfrac{\theta}{2}$, solve $2\cos\theta + 3\sin\theta = -2$ for $0 \le \theta \le 2\pi$.

Hint: compare the coefficient of $\cos\theta$ with the constant term before you start.

PP 3 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q12(b)** 3 marks

Using the t-formula solve the equation $\dfrac{5\sin x}{2} + \cos x = 2$ for $0° \le x \le 180°$. Answer correct to the nearest minute.

Hint: multiply by 2 first. With $0° \le x \le 180°$, what is the range of $\frac{1}{2}x$?

PP 4 **Normanhurst Boys' High School, 2021, Mathematics Extension 1 (Year 12 Course) Assessment Task 4 (Trial Examination), Q11(c)** 3 marks

Using the substitution $t = \tan\theta$, solve for $0° \leq \theta \leq 360°$:

$$2\sin 2\theta = 2 + 4\cos 2\theta$$

Hint: $\sin 2\theta$ and $\cos 2\theta$ in terms of $t = \tan\theta$. Then check $\theta = 90°$ and $270°$.

PP 5 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q14(a)** 3 marks

Using the substitution $t = \tan x$, solve $3\sin 2x - 5\cos 2x = 5$ for $0 \leq x \leq 2\pi$.

Hint: watch what happens to the t^2 terms, and think about which values of x the substitution cannot reach.

PP 6 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(a)** 4 marks

Use the t-formulae to solve the equation $\sin 4\theta + \sqrt{3}\cos 4\theta = 0$ for $0 \leq \theta \leq \pi$.

Hint: let $t = \tan 2\theta$, so $\sin 4\theta = \dfrac{2t}{1+t^2}$. Write down the range of 2θ.

PP 7 **Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q12(b)** 3 marks

It is given that $t = \tan A$ and $p\cos 2A - \sin 2A = 1$, where $p \in \mathbb{R}$. Show that

$$t = \frac{p-1}{p+1}$$

where $0 \leq A \leq \frac{\pi}{2}$.

Hint: you will get a quadratic in t. One root is a simple number; use the domain of A to reject it.

CHAPTER 11 REVIEW

Chapter review exercise

Mixed practice from the whole chapter. Decide which method fits before you start each question.

1 Solve, for $0 \leq x \leq 2\pi$:

a $\sin 2x + \sin x = 0$

b $\cos 2x + \cos x = 0$

c $\cos 2x + 5\sin x + 2 = 0$

d $2\sin\left(x - \frac{\pi}{6}\right) = \cos\left(x - \frac{\pi}{3}\right)$

2 a Express $\sin x - \cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

b Hence solve $\sin x - \cos x = \sqrt{2}$, for $0 \leq x \leq 2\pi$.

3 a Express $\sqrt{3}\cos x + \sin x$ in the form $A\cos(x - \theta)$, where $A > 0$ and $0 < \theta < \frac{\pi}{2}$.

b Hence solve $\sqrt{3}\cos x + \sin x = -1$, for $0 \leq x \leq 2\pi$.

4 a Express $2\sin x + \sqrt{5}\cos x$ in the form $R\sin(x+\alpha)$, where $R > 0$ and α is acute.

b Hence solve $2\sin x + \sqrt{5}\cos x = 3$, for $0° \leq x \leq 360°$, writing the solution in degrees correct to one decimal place.

5 a Express $3\cos x - 2\sin x$ in the form $A\cos(x+\theta)$, where $A > 0$ and θ is acute.

b Hence solve $3\cos x - 2\sin x = 1$, for $0° \leq x \leq 360°$, writing the solutions correct to the nearest minute.

6 Use a suitable t-formula to solve $\sin x = \tan \frac{1}{2}x$, for $0 \leq x \leq 2\pi$.

7 Consider the equation $7\sin x + \cos x = 5$.

a Show that the equation can be written as $3t^2 - 7t + 2 = 0$, where $t = \tan \frac{x}{2}$.

b Hence show that the equation has solutions $x = 2\tan^{-1}\frac{1}{3}$ and $x = 2\tan^{-1}2$.

8 Use t-formulae to solve $4\sin x - 2\cos x = 3$, for $0 \leq x \leq 2\pi$. Write the solutions correct to 2 decimal places.

9 a Use compound- and double-angle formulae to prove that $\cos 3x = 4\cos^3 x - 3\cos x$.

b Hence solve $\cos 3x + \sin 2x + \cos x = 0$, for $0 \leq x \leq 2\pi$.